Sai Charan Reddy

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Summary

Computer Science student specializing in Artificial Intelligence, with hands-on experience building AI-powered applications using Machine Learning, Generative AI, NLP, and LLMs. Skilled in Python, PyTorch, LangChain, LangGraph, RAG, and prompt engineering, with practical experience in transformer architectures, hybrid retrieval, multi-agent systems, and vector databases. Strong analytical and problem-solving skills, focused on developing scalable AI solutions.

Education

Bennett University, Greater Noida
Bachelor of Technology in Computer Science — *Sep 2023 – Present*

Technical Skills

Programming Languages: Python, Java
AI/ML & Generative AI: Machine Learning, Deep Learning, NLP, LLMs, Generative AI, RAG, Prompt Engineering, Feature Engineering, Hyperparameter Tuning, Model Development & Deployment
Frameworks & Libraries: LangChain, LangGraph, FastAPI, NumPy, Pandas, Matplotlib, Seaborn
LLMs & AI Platforms: Gemini, Ollama
Databases: PostgreSQL, Qdrant (Vector Database)
Tools & Platforms: Git, Docker, LangSmith, n8n, Antigravity
Core Concepts: Data Structures & Algorithms

Projects

SkillLens – AI Resume Screening — **GitHub**
Python, FastAPI, Gemini, BAAI/bge-base-en-v1.5, SPLADE++ — *2026 – Present*

- Built an AI-powered resume screening pipeline using hybrid retrieval with dense (BAAI/bge-base-en-v1.5) and sparse (SPLADE++) embeddings to identify relevant candidates beyond keyword matching.
- Developed a resume-level ranking algorithm that averages each resume's top-3 chunk similarity scores, reducing score dilution across long documents and surfacing the most relevant sections.
- Integrated Gemini to enhance natural-language queries for retrieval and automatically infer the requested number of results from user phrasing, reducing manual configuration.

Float Chat – Multi-Agent Oceanographic Data Retrieval — **GitHub**
Python, LangGraph, Gemini, MongoDB, PostgreSQL — *2025*

- Developed an oceanographic data retrieval system combining MongoDB geospatial filtering and metadata retrieval with PostgreSQL structured queries, using spatial indexing instead of relying solely on semantic similarity search.
- Architected a three-agent LangGraph workflow comprising Decomposition, Filter, and SQL agents to route user queries to appropriate data retrieval operations.
- Integrated MongoDB for location-based and metadata filtering and PostgreSQL for structured oceanographic float-profile retrieval.

GPT-2 Implementation from Scratch — **GitHub**
Python, PyTorch, Transformers, Deep Learning — *2025*

- Implemented the GPT-2 architecture from scratch in PyTorch, including token and positional embeddings, causal masked multi-head self-attention, and pre-norm transformer blocks with residual connections.
- Implemented KV-cache support with separate prefill and decode passes for autoregressive generation; trained on FineWeb using cosine learning rate decay, gradient accumulation, and bfloat16 mixed precision, achieving a validation loss of 2.77.
- Explored transformer internals, attention mechanisms, positional encoding, and training dynamics to build a deeper understanding of the architecture underlying large language models.

Certifications

- Foundations of Deep Learning: Concepts and Applications — **Link**
- Data Structures — **Link**
- AI Workflow: Data Analysis and Hypothesis Testing — **Link**